

Conditioning tensors

Base origin context

$$q_B: B \times 46$$

Future calendar

$$C^+: B \times 24 \times 9$$

Residual neighbours

$$\Delta_G: B \times 8 \times 24$$

Reliability inputs

$$z_t: B \times 18$$

Trust-gated residual correction

Context XGBoost
24 horizon heads

$$\hat{Y}_X: B \times 24$$

Context LightGBM
24 horizon heads

$$\hat{Y}_L: B \times 24$$

base

$$0.3479 \hat{Y}_X + 0.6521 \hat{Y}_L$$

Robust residual
alignment

$$r_t: B \times 24$$

Depth-2 trust gate

$$g_t: B \times 1$$

+

$$\hat{Y}_{TRACE} = \hat{Y}_B + g_t r_t$$

DFEH evidence output

Base feature effects
top fields

Residual window IDs
top 8 analogues

Linked event IDs
prior 72 h

Gate, drift, uncertainty
machine-readable

2,460 evidence records

forecast